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Article

Population-Weighted Exposure Assignment from Residential Postal Codes: a National Comparison of Methods

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Abstract

The Postal Code Conversion File Plus (PCCF+) assigns a residential postal code with several candidate locations to one location drawn at random in proportion to population. Two non-random alternatives that use the same candidates and weights, (A) the weighted mean of the exposure across candidates and (B) the exposure at the weighted centroid, were compared with (C) PCCF+ random allocation, using (D) the exposure at the address as the reference, for 12,410,638 residential addresses in the National Address Register. Exposures were greenness (MODIS Enhanced Vegetation Index) and 2021 Material and Social Deprivation Index quintiles. The weighted mean had the same greenness bias as PCCF+, and its mean absolute error was 13.5% lower overall and 20% lower for rural postal codes. The error of random allocation decomposed exactly into the error of the weighted mean and an allocation variance, which accounted for 34% of its mean squared error for greenness and 49% for the material deprivation score among postal codes with six-, five- or four-character weighted links. The weighted centroid had the smallest positional error but did not improve greenness. The quintile of the weighted mean score compressed assignments toward the middle quintiles, whereas the modal candidate quintile reduced material quintile misclassification from 12.2% to 10.2% overall and from 46.2% to 37.0% for rural postal codes. Differences were small for urban letter-carrier postal codes. The weighted mean for continuous exposures and the modal quintile for area-based quintiles are recommended in place of random allocation, and a validated R function is provided.

Keywords: geocoding; postal codes; population weighting; exposure assessment; Berkson error; greenness; area-based deprivation; rural health; national address register; Canada

1. Introduction

Residential postal codes are a common link between Canadian health records and the environmental and social conditions measured around a person's home. Administrative health data, cohort studies and survey files often retain the six-character postal code but not the civic address, so exposures such as greenness, air pollution or area-based deprivation are attached through a postal code conversion step [1,2]. A postal code is a mail delivery construct rather than a location. An urban postal code served by a letter carrier usually corresponds to one side of a street block or a single building, whereas a rural postal code can serve a post office and its surrounding territory over many kilometres [3]. The Postal Code Conversion File Plus (PCCF+) distributed by Statistics Canada manages this ambiguity by assigning each record with a multi-location postal code to one candidate location, selected at random in proportion to the census population of that postal code in each candidate dissemination area (DA) [4].

Earlier Canadian evaluations found that the accuracy of postal code geocoding depends mainly on delivery mode, with median errors of a few hundred metres in larger communities and several

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kilometres in rural areas [5–8]. A recent national validation compared PCCF+ Version 8D representative points with building-level reference locations for 12.4 million residential addresses in the National Address Register (NAR) [9]. It reported a median positional error of 65 m for urban houses and 3.1 km for rural postal codes, an upward bias in greenness assigned to rural postal codes, and misclassification of the material deprivation quintile for 46% of rural postal-code addresses. Because the errors were concentrated among rural residents, they are differential across the rural-urban gradient and can distort comparisons of health inequalities between and within settings [9].

When a postal code links to several candidate locations, the representative point selected carries no information about where that person lives. Under random allocation, the variation in exposure values among the candidate points therefore enters the assigned exposure as noise. In measurement error terms, this noise behaves like classical error, which tends to attenuate exposure-response associations [10,11]. An alternative is to assign each postal code the population-weighted mean of the exposure across *all* of its candidate locations. The assigned values from this method thus approximate the expected exposure of residents of the postal code. Thus, the remaining error is closer to Berkson error, which leaves slopes approximately unbiased in linear models at the cost of precision [10–12]. This method also assigns the same value to every record with a given postal code, which makes the results reproducible without a random seed, unlike in the random allocation method where records with the same postal code may have different assigned exposure values.

The authors have previously applied population-weighted averaging of this kind to link environmental exposures to postal codes in a cohort study that included mixed urban and rural respondents [13,14]. The Canadian Urban Environmental Health Research Consortium (CANUE), a national exposure platform, distributes exposure values indexed to postal codes [15]. Postal code linked exposure data have been used in numerous studies across Canada, including in studies examining greenness and social deprivation [16–20].

However, averaging methods have not been compared with PCCF+ random allocation against a national address register, and the conditions under which averaging helps or harms have not been described. Averaging is expected to help most for smoothly varying exposures such as greenness. For categorical area-based measures, such as deprivation quintiles, averaging scores across heterogeneous areas may produce values that do not occur in any candidate area and may pull assignments toward the middle of the distribution.

This study compared four methods for assigning exposures to residential postal codes:

A. the population-weighted mean of exposures across all candidate locations;
B. assignment at the population-weighted geographic centroid of the candidate locations;
C. PCCF+ population-weighted random allocation; and,
D. the exact address location, which served as the reference.

The objectives were:

1) to describe the candidate locations and weights for Canadian residential postal codes by link source, delivery mode and rurality;
2) to compare the positional error of the four methods;
3) to compare the agreement of greenness and area-based deprivation assigned by the weighted mean, weighted centroid and PCCF+ with values at the address; and,
4) to decompose the error of random allocation into the error of the weighted mean and an allocation variance.

A further objective was to provide a portable, validated and reproducible R implementation of the weighted mean and weighted centroid methods. Code for PCCF+ and the address reference was released with an earlier validation study [9].

Four hypotheses were posited for this analysis. First, the weighted mean and PCCF+ would have similar greenness bias, but the weighted mean would have smaller MAE and RMSE, with gains concentrated in rural delivery modes and in six-, five- and four-character weighted links. Second, the weighted mean would reduce material and social quintile misclassification relative to PCCF+ for rural delivery modes but would assign fewer addresses to the extreme quintiles. Third, the weighted

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centroid would have the smallest positional error but would not consistently outperform the weighted mean for greenness, because the centroid of dispersed rural candidates may fall away from settled areas. Fourth, the methods would differ little for urban letter-carrier postal codes, where most postal codes have a single candidate or candidates within the same DA.

2. Materials and Methods

2.1. Study Design and Data Sources

This cross-sectional methods comparison used the address sample, reference locations, PCCF+ output and exposure layers assembled for the national PCCF+ validation study [9], so that PCCF+ and the address reference were identical to those reported previously and the new comparisons isolate the effect of the assignment method. The NAR (June 2026 release) provided civic addresses, building use, mailing postal codes, building points and blockface representative points [21]. The Open Database of Addresses (ODA, version 1) supplied alternative reference locations where a NAR building point was absent [22]. PCCF+ Version 8D (December 2025 postal codes, 2021 census geography) supplied the candidate locations and population weights [4]. Greenness was measured with the MODIS Terra vegetation index product MOD13Q1 Version 6.1 [23,24], and area-based deprivation with the 2021 Material and Social Deprivation Index (MSDI) of the *Institut national de santé publique du Québec* [25,26]. Rurality was classified by the 2021 Statistical Area Classification (SAC) assigned through the 2021 Geographic Attribute File [27,28]. All distance and centroid calculations used the Statistics Canada Lambert projection (EPSG:3347), and DA assignment used the 2021 DA cartographic boundary file.

2.2. Residential Addresses and Reference Locations

The analytic sample was the primary sample of the validation study: 12,412,300 residential NAR addresses with a valid six-character mailing postal code, a reference location and a PCCF+ assignment, excluding urban letter-carrier addresses whose reference location was a NAR blockface point [9]. Addresses whose postal code PCCF+ resolved only from three or fewer characters (three-character weighted links, n = 1,187; two-character links, n = 475) had no candidate location (Section 2.3) and were excluded, leaving 12,410,638 addresses in 746,916 postal codes. The reference location was the NAR building point (94.1%), a matched ODA point (2.0%) or the NAR blockface point (3.9%). Each address kept its PCCF+ delivery mode type (DMT), link source, representative geographic point, and DA from the validation study, and its reference DA, SAC type, and province. The weighted mean and weighted centroid methods were calculated for each postal code, and addresses were the unit of comparison, so each postal code contributed in proportion to its number of residential addresses.

2.3. Candidate Locations and Population Weights

For each postal code p, the candidate locations are the PCCF+ records from which PCCF+ makes its random selection, and the PCCF+ link source reported for the postal code determines which file supplies them (Table 1). First, residential postal codes served by rural routes, suburban services or post office boxes are matched using the six-character weighted conversion file (WCF), giving six-character weighted links (Link_Source C), where each candidate is a DA-representative point with a population weight. Remaining postal codes are then matched to either the file of unique links (Link_Source F) or the file of duplicate links (Link_Source D), in which a postal code links to several blockface, dissemination block or DA representative points. Postal codes that cannot be matched on all six characters are then imputed from the five- or four-character weighted conversion files, giving five- and four-character weighted links (Link_Source 5 and 4). For postal codes matched only on their first three characters (three-character weighted links, Link_Source 3), PCCF+ draws a record from the three-character WCF but then replaces its DA with an unknown code and does not retain the

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geographic location, so these records, like those assigned only partial geography (two-character and lower, Link_Source 2, 1 and 0), have no representative point location and are thus excluded here [4].

Table 1. Candidate locations and weights by PCCF+ link source.

Link source	PCCF+ file	Candidate location	PCCF+ 8D selection	Weight in methods A and B
Unique (F)	pccfuniq	Single representative point	No selection	1
Six-character weighted (C)	wc6dups	DA representative point	Random, PC6DAWT	PC6DAWT
Duplicate (D)	pccfdups	Blockface, block or DA point	Random, equal	Equal; DA population share (sensitivity)
Five-character weighted (5)	wc5dups	DA representative point	Random, PC5DAWT	PC5DAWT
Four-character weighted (4)	wc4dups	DA representative point	Random, PC4DAWT	PC4DAWT
Three-character weighted (3)	wc3dups	None retained	Random, PC3DAWT	Excluded
Two-character or lower (2, 1, 0)	none	No coordinates	Not applicable	Excluded

DA, dissemination area. PC6DAWT to PC4DAWT are the census population weights of the PCCF+ weighted conversion files for six- to four-character postal codes, and PC3DAWT that of the three-character file, whose draw PCCF+ 8D does not retain. PCCF+ also imputes a dissemination block code within the assigned DA, but the coordinates remain those of the candidate record.

Let $j = 1, \ldots, m_p$ index the candidate records of postal code p, with projected coordinates (x_{pj}, y_{pj}), DA $a(j)$ and raw weight v_{pj}. The normalized weight is

$$w_{pj} = \frac{v_{pj}}{\sum_{k=1}^{m_p} v_{pk}} \quad (1)$$

For six-, five- and four-character weighted links, v_{pj} is the PCCF+ population weight of the candidate. For duplicate links, PCCF+ selects among the duplicate records with equal probability [4], and the primary analysis used the same equal weights ($v_{pj} = 1$), so that the weights of the weighted mean and weighted centroid methods are the PCCF+ selection probabilities for every link source. An alternative for duplicate links, examined in a sensitivity analysis, uses the 2021 population share of each candidate DA, divided equally among the candidate records within that DA, so that a DA represented by several blockface points is not given more weight than its population warrants:

$$v_{pj} = \frac{P_{a(j)}}{n_{p,a(j)}} \quad (2)$$

where P_a is the 2021 census population of DA a and $n_{p,a}$ is the number of candidate records of postal code p in DA a. Where every raw weight of a postal code was zero or missing, which occurs for postal codes whose candidate DAs have no enumerated population, equal weights were used, consistent with the fallback applied to the PCCF+ run in the validation study [9]. The number of candidates, the effective number of candidates ($1/\Sigma w_{pj}^2$), and the number of distinct candidate DAs were recorded for each postal code.

2.4. Exposure Assignment Methods

Greenness exposure was the growing-season Enhanced Vegetation Index (EVI), calculated for each 231.7 m MODIS cell as the mean of 16-day composites beginning between 1 June and 31 August with valid quality flags, averaged over 2025 and 2026, as in the validation study [9,23]. EVI was extracted for the containing cell (pixel), and for the mean of 3 × 3 (approximately 700 m, primary

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measure) and 5 × 5 (approximately 1.2 km) windows. Extraction by containing cell is equivalent to assigning each point the value of the nearest MODIS cell centre.

Deprivation was represented by the national material and social factor scores and quintiles of the MSDI for 2021 DAs [26]. As each candidate record includes a DA identifier, the MSDI was linked to postal codes by the assigned DA identifier rather than by a spatial join. Let e_{pj} denote the exposure at candidate j, S_a the MSDI factor score of DA a, and Q_a its quintile.

Method A (weighted mean).

The greenness of postal code p is the weighted mean of the candidate values,

$$\hat{e}_p^A = \sum_{j=1}^{m_p} w_{pj}\ e_{pj} \tag{3}$$

with weights renormalized over candidates with a non-missing value, and the share of weight with missing EVI recorded. For deprivation, the weighted mean of the factor scores is converted to a quintile with the national quintile cut points $c_1 < c_2 < c_3 < c_4$ of the MSDI,

$$\hat{S}_p = \sum_{j=1}^{m_p} w_{pj}\ S_{a(j)}, \quad \hat{q}_p^A = 1 + \sum_{k=1}^{4} \mathbf{1}\left[\hat{S}_p > c_k\right] \tag{4}$$

The cut points were recovered from the MSDI table as the midpoint between the largest score in quintile k and the smallest score in quintile $k + 1$, and reproduced the published quintile of every DA. The weighted distribution of candidate quintiles was also retained,

$$\pi_p(q) = \sum_{j=1}^{m_p} w_{pj}\ \mathbf{1}\left[Q_{a(j)} = q\right], \quad q = 1, \ldots, 5 \tag{5}$$

which gives, for each postal code, the probability that a randomly allocated record falls in each quintile and links the weighted mean to the postal-code correction developed in the validation study [9]. A second quintile rule was also applied: the modal quintile, the value of q that maximizes $\pi_P(q)$, which minimizes the expected misclassification of a postal code given its candidate distribution. Ties were resolved in favour of the quintile closest to the quintile of the weighted mean score, and then the lower quintile.

Method B (weighted centroid).

Each postal code receives a single point at the weighted centroid of its candidates in projected coordinates,

$$x_p^B = \sum_{j=1}^{m_p} w_{pj}\ x_{pj}, \quad y_p^B = \sum_{j=1}^{m_p} w_{pj}\ y_{pj} \tag{6}$$

EVI was extracted at this point, and MSDI was attached from the DA containing it by point-in-polygon. A postal code with a single candidate kept the DA of that candidate, and centroids that fell outside every DA, for example in water, were assigned to the nearest DA and flagged.

Method C (PCCF+).

Each address i in postal code p receives the candidate J_i drawn by PCCF+ 8D,

$$\Pr(J_i = j) = w_{pj}^P, \quad e_i^C = e_{pJ_i} \tag{7}$$

where w^P_{pj} are the PCCF+ selection probabilities (Table 1). The realized PCCF+ point, DA, EVI and MSDI values were taken from the validation study.

Method D (address reference).

Each address receives the EVI of its reference location and the MSDI score and quintile of the DA containing the reference location, e_i^D and t_i.

For unique links, the weighted mean, weighted centroid and PCCF+ give the same value. When the weights of the weighted mean and PCCF+ are the same, which under the primary weights is the

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case for every link source, the weighted mean is the expected value of PCCF+ over the random draw, and the mean squared error (MSE) of PCCF+ decomposes as

$$\mathrm{E}[(e_i^C - e_i^D)^2] = \left(\hat{e}_p^A - e_i^D\right)^2 + V_p, \quad V_p = \sum_{j=1}^{m_p} w_{pj}\left(e_{pj} - \hat{e}_p^A\right)^2 \tag{8}$$

where the expected value is over the draw and V_p is the allocation variance of a given postal code p. The weighted mean therefore cannot have a larger expected squared error than PCCF+, and the size of the reduction is the mean allocation variance.

Table 2. Summary of the four assignment methods.

Method	Location used	Greenness	Deprivation	Random
A. Weighted mean	All candidates	Weighted mean of candidate EVI	Quintile of weighted mean score; modal candidate quintile	No
B. Weighted centroid	One computed point	EVI at centroid	Score and quintile of centroid DA	No
C. PCCF+	One drawn candidate	EVI at drawn point	Score and quintile of drawn DA	Yes
D. Address	Reference location	EVI at reference point	Score and quintile of reference DA	No

EVI, Enhanced Vegetation Index. Method D is the reference standard.

2.5. Outcomes

Positional error. For the weighted centroid and PCCF+, positional error was calculated as the geodesic distance between the assigned point and the reference location, as used in the validation study [9]. The weighted mean does not assign a single point, so its positional error is defined as the expected distance from the reference location r_i to all candidate locations,

$$\hat{d}_i^A = \sum_{j=1}^{m_p} w_{pj}\; d\left(r_i, x_{pj}\right), \quad d\left(r_i, x_p^B\right) \le \hat{d}_i^A \tag{9}$$

where d is the distance (Lambert projection) and the inequality follows from the triangle inequality. Under PCCF+ weights, the expected distance is also the expected positional error of PCCF+. DA agreement was reported for the weighted centroid and PCCF+; for the weighted mean, it is reported as the total weight of the candidates located in the reference DA.

Greenness agreement. For each method M in {A, B, C}, the difference $e^M - e^D$ was summarized by the mean (bias), mean absolute error (MAE), root mean square error (RMSE) and the Pearson correlation between assigned and reference values. The paired difference in absolute error between the weighted mean and PCCF+,

$$\Delta_i = \left|\hat{e}_p^A - e_i^D\right| - \left|e_i^C - e_i^D\right| \tag{10}$$

was the primary comparison, with negative values favouring the weighted mean. The same paired difference was computed for the weighted centroid against PCCF+ and for the weighted mean against the weighted centroid.

Deprivation agreement. For each method, misclassification was any disagreement between the assigned quintile and the reference quintile t_i. This was summarized by the misclassification rate, the proportion assigned a higher or lower quintile, the proportion misclassified by two or more quintiles, and linearly weighted kappa as defined in the validation study [9]. As averaging may compress quintile assignment toward the middle of the distribution, the shares of addresses assigned to quintiles 1 and 5 were compared with the reference distribution. The expected misclassification of PCCF+ for an address, $1 - \pi_p(t_i)$, was compared with its realized misclassification.

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Candidate heterogeneity. For each postal code, the weighted standard deviation of candidate EVI (the square root of V_p), the number of distinct candidate quintiles, and the largest $\pi_p(q)$ are used to identify where the methods can differ.

2.6. Statistical Analysis

All outcomes were described overall and by link source, delivery mode, rurality, their combination, and province, using the delivery mode and SAC groupings of the validation study [9]. Delivery mode was grouped as urban letter carrier to a house (DMT A) or apartment (B), business building or large volume receiver (E, G), rural postal code (W), rural route or suburban service (H, T), and post office box, general delivery or other (J, K, M, R, X). Rurality was grouped as census metropolitan area (CMA), census agglomeration (CA), strong, moderate, weak and no metropolitan influenced zone (MIZ), and territories.

Uncertainty was estimated with a Bayesian bootstrap over postal codes [29], with 200 replicates, because addresses within a postal code share candidate sets and, for the weighted mean and weighted centroid, assigned values. Intervals are reported for the mean paired differences Δ, for differences in misclassification rate between methods, and for differences in weighted kappa. Two models were fitted on cell-level statistics, as in the validation study: an ordinary least squares model for Δ with delivery mode, rurality and their interaction, and a logistic model for material quintile misclassification with method, delivery mode, rurality and their interactions, both adjusted for province. Cells with fewer than 100 addresses were not modelled. Given the size of the sample, interpretation focuses on effect sizes rather than statistical significance.

2.7. Sensitivity Analyses

Six sensitivity analyses were conducted: (1) DA population-share weights for duplicate links (Equation 2); (2) weighted centroids calculated from latitude and longitude rather than projected coordinates, as in earlier applications of the method; (3) the pixel and 5 × 5 EVI windows and each year separately; (4) restriction to addresses with a NAR building reference point; (5) inclusion of urban letter-carrier addresses with a blockface reference point; and (6) a check of MSE decomposition (Equation 8) for every EVI measure and both MSDI scores, in which the observed MSE of PCCF+ was compared with the sum of the MSE of the weighted mean and the mean allocation variance for six-, five- and four-character weighted links and for duplicate links.

2.8. Portable Implementation

An R function implementing the weighted mean and weighted centroid for any dataset with six-character postal codes was written and validated using the source data. Inputs for this function are a table of records with postal codes, the data directory of a licensed PCCF+ 8D installation, an exposure layer given as a raster (extracted by containing cell) or as points (assigned by nearest feature), and optional DA-level attributes such as MSDI scores and quintiles. The link source of each postal code is taken from a PCCF+ run where one exists; otherwise it is inferred with the matching sequence of the PCCF+ 8D residential routine, including the recoding of postal codes with rural route, general delivery, post office box and similar delivery modes to the three-character level. Postal codes that PCCF+ would resolve from three or fewer characters are returned without values.

For each postal code, the function returns the link source, the number and effective number of candidates, the weighted mean exposure, weighted standard deviation and share of weight with missing values, the weighted mean DA attributes with their quintile, quintile probabilities and modal quintile, and the weighted centroid with its exposure and DA attributes. Agreement with this analysis was tested for all postal codes in Prince Edward Island, and the inferred link sources were compared with those reported by PCCF+ for every postal code in the national sample.

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2.9. Software, Reproducibility and Use of Artificial Intelligence Tools

Analyses were conducted in R [30] with the data.table [31], sf [32], terra [33] and ncdf4 [34] packages. Analysis code and the reproducible R function are available via GitHub, excluding PCCF+ files, which are licensed by Statistics Canada. Artificial intelligence tools (Claude Code, Anthropic) were used to develop reproducible code for dissemination, to validate analytic results, and to edit the manuscript for clarity and grammar.

3. Results

3.1. Candidate Locations

The 12,410,638 addresses from the NAR were in 746,916 postal codes (Table 3). More than half of the addresses (55.9%) had a unique link, a quarter (24.8%) a duplicate link, and 19.2% a six-character weighted link; fewer than 3,000 addresses had five- or four-character weighted links. The average address had 2.69 candidate locations, or 2.27 effective candidates after weighting, but the candidate sets differed by link source and delivery mode. Addresses with six-character weighted links had on average 7.4 candidates (5.2 effective), 94.1% had candidates in more than one DA, and 86.9% had candidates with more than one material deprivation quintile, with a mean largest quintile probability of 0.64. Rural postal codes were similar, with 7.8 candidates and a weighted standard deviation of candidate EVI of 0.056, compared with 0.004 for urban houses and less than 0.001 for apartments. Candidates of duplicate links were more homogeneous, with a mean candidate EVI standard deviation of 0.010 and more than one quintile for 19.7% of addresses.

Table 3. Candidate locations by PCCF+ link source and postal code delivery mode type.

Group	Postal codes	Addresses	%	Cand.	Eff. cand.	Multi-DA (%)	SD EVI	Multi-quintile (%)	Largest π
All addresses	746,916	12,410,638	100.0	2.69	2.27	26.1	0.013	21.6	0.91
Link source									
Unique (F)	529,901	6,942,142	55.9	1.00	1.00	0.0	0.000	0.0	1.00
Six-character weighted (C)	9,435	2,387,883	19.2	7.38	5.21	94.1	0.055	86.9	0.64
Duplicate (D)	207,365	3,077,779	24.8	2.86	2.86	32.3	0.010	19.7	0.92
Five-character weighted (5)	108	740	0.0	3.70	2.73	82.2	0.029	78.0	0.69
Four-character weighted (4)	107	2,094	0.0	10.03	5.45	100.0	0.051	99.6	0.57
Delivery mode									
Urban LC: house	713,057	7,989,501	64.4	1.69	1.68	12.5	0.004	7.7	0.97
Urban LC: apartment	25,180	2,052,235	16.5	1.08	1.08	1.3	0.000	0.6	1.00
Business / LVR	766	30,229	0.2	1.07	1.07	0.7	0.000	0.3	1.00
Rural postal code	4,941	2,147,208	17.3	7.75	5.54	95.3	0.056	88.0	0.63
Rural route / suburban	826	155,388	1.3	5.42	3.22	95.3	0.049	90.8	0.67
PO box / general delivery	1,840	31,953	0.3	5.30	4.00	65.6	0.033	58.9	0.76
Unknown / retired	333	4,124	0.0	6.17	3.66	66.2	0.032	65.2	0.72

Values are address-weighted means or percentages of addresses. Cand., number of PCCF+ candidate records; Eff. cand., effective number of candidates, $1/\Sigma w^2$ under the primary weights; Multi-DA, candidates in more than one DA; SD EVI, weighted standard deviation of the 3 × 3 EVI (2025–2026) across candidates; Multi-quintile, candidates in more than one material deprivation

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quintile; largest π, largest weighted share of one material quintile. LC, letter carrier; LVR, large volume receiver. Five- and four-character weighted links account for less than 0.1% of addresses. By province: Supplementary Table S1.

3.2. Positional Error

The weighted centroid had the smallest positional error overall, with a median of 78 m and a mean of 1,110 m. This compared with an expected distance of 87 m (median) and 1,433 m (mean) for the weighted mean and a median of 81 m and mean of 1,424 m for PCCF+ (Table 4). The expected planar distance of the weighted mean (which equals the expected positional error of PCCF+) matched the realized planar error of PCCF+ (mean of 1,433 m vs. 1,435 m). For urban letter-carrier postal codes, the median errors of the three methods differed by less than 10 m.

For rural postal codes, the median error was 2.3 km for the centroid, 3.1 km for PCCF+ and 4.2 km for the expected distance. However, while the median error in distance was large, the DA agreement was similar across the three methods (28.3% to 29.6%). In the Territories and areas with weak or no MIZ, the weighted centroid had a larger median error than PCCF+ (for no MIZ, 2.9 km vs. 2.2 km) and lower DA agreement (39.4% vs. 54.8%), which is consistent with a pattern where centroids fall between widely separated settlements.

Table 4. Positional error (m) and same dissemination area agreement (%) by method, delivery mode and rurality.

Group	Addresses	Median A	Median B	Median C	Mean A	Mean B	Mean C	Same DA A	Same DA B	Same DA C
All addresses	12,410,638	87	78	81	1,433	1,110	1,424	80.6	80.4	80.6
Delivery mode										
Urban LC: house	7,989,501	72	63	65	175	159	173	91.8	91.9	91.8
Urban LC: apartment	2,052,235	54	53	53	97	95	96	94.7	94.4	94.7
Business / LVR	30,229	48	48	48	130	128	130	92.2	92.0	92.2
Rural postal code	2,147,208	4,202	2,344	3,125	6,961	5,277	6,926	29.5	28.3	29.6
Rural route / suburban	155,388	4,399	2,947	3,762	5,294	3,941	5,205	37.0	37.3	36.9
PO box / general delivery	31,953	3,307	2,321	2,429	12,288	10,923	12,330	22.0	20.5	21.8
Unknown / retired	4,124	2,438	2,350	2,434	5,188	4,645	5,107	12.2	13.1	11.9
Rurality										
CMA	8,761,718	69	62	63	334	261	330	89.6	89.7	89.6
CA	1,413,322	92	82	84	975	739	963	81.1	80.9	81.2
Strong MIZ	739,531	3,376	1,783	2,367	4,721	3,480	4,685	39.5	39.4	39.6
Moderate MIZ	931,063	3,440	1,714	2,093	7,226	5,838	7,198	46.8	46.4	46.9
Weak MIZ	510,939	2,655	1,189	1,046	5,550	4,094	5,561	48.2	45.7	48.3
No MIZ	44,110	4,597	2,913	2,218	7,983	6,391	8,008	54.9	39.4	54.8
Territories	9,955	1,828	1,341	990	6,826	6,081	6,879	49.3	39.5	49.3

A, weighted mean (expected distance from the address to the candidates; no single point is assigned); B, weighted centroid; C, PCCF+. Distances are geodesic for B and C and planar (Lambert projection) for A. Same DA: for A, the total weight of candidates in the reference DA; for B and C, the share of addresses assigned the reference DA. LC, letter carrier; LVR, large volume receiver; CMA, census metropolitan area; CA, census agglomeration; MIZ, metropolitan influenced zone. By province and link source: Supplementary Tables S2 and S5.

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3.3. Greenness

The weighted mean had the same greenness bias as PCCF+ in every group with more than a few thousand addresses, an overall bias of 0.0068 EVI units and 0.035 for six-character weighted links (Table 5). This was expected because averaging removes variation among candidates but not any systematic difference between the candidate values and the realized addresses, such as DA representative points that lie in greener surroundings than the dwellings they represent. The MAE of the weighted mean was 13.5% lower than that of PCCF+ (0.0217 vs. 0.0251), its RMSE 17% lower (0.0438 vs. 0.0527), and its correlation with address values higher (0.922 vs. 0.892). The mean paired difference in absolute error was −0.0034 (95% interval −0.0036 to −0.0033) overall and −0.0160 (−0.0167 to −0.0155) for six-character weighted links, where the MAE fell by 20% and the correlation rose from 0.533 to 0.654. The gain was smaller for duplicate links (−0.0015) and negligible for urban letter-carrier postal codes (−0.0006 for houses and less than 0.0001 for apartments).

Table 5. Agreement of greenness (EVI, 3 × 3 window, 2025–2026) with the address value by method, link source and delivery mode.

Group	Addresses	Bias A	Bias C	MAE A	MAE B	MAE C	r A	r C	Δ_{AC} (95% interval)
All addresses	12,396,809	0.0068	0.0068	0.0217	0.0252	0.0251	0.922	0.892	−0.0034 (−0.0036, −0.0033)
Link source									
Unique (F)	6,941,466	0.0001	0.0001	0.0097	0.0097	0.0097	0.972	0.972	0.0000 (0.0000, 0.0000)
Six-character weighted (C)	2,378,363	0.0348	0.0348	0.0651	0.0824	0.0811	0.654	0.533	−0.0160 (−0.0167, −0.0155)
Duplicate (D)	3,074,147	0.0004	0.0004	0.0151	0.0157	0.0166	0.956	0.936	−0.0015 (−0.0015, −0.0014)
Five-character weighted (5)	739	0.0411	0.0384	0.0808	0.0831	0.0861	0.406	0.410	−0.0053 (−0.0129, 0.0026)
Four-character weighted (4)	2,094	0.0714	0.0716	0.0916	0.1324	0.1030	0.332	0.265	−0.0114 (−0.0169, −0.0069)
Delivery mode									
Urban LC: house	7,988,160	0.0004	0.0004	0.0119	0.0121	0.0125	0.965	0.957	−0.0006 (−0.0006, −0.0006)
Urban LC: apartment	2,052,090	−0.0002	−0.0002	0.0099	0.0100	0.0100	0.968	0.966	0.0000 (−0.0001, 0.0000)
Business / LVR	30,229	0.0006	0.0005	0.0094	0.0093	0.0095	0.978	0.977	−0.0001 (−0.0004, 0.0000)
Rural postal code	2,134,977	0.0358	0.0359	0.0656	0.0837	0.0821	0.642	0.514	−0.0165 (−0.0172, −0.0160)
Rural route / suburban	155,286	0.0343	0.0345	0.0681	0.0822	0.0821	0.477	0.369	−0.0140 (−0.0157, −0.0125)
PO box / general delivery	31,944	0.0083	0.0083	0.0615	0.0714	0.0707	0.675	0.613	−0.0093 (−0.0112, −0.0080)
Unknown / retired	4,123	0.0404	0.0401	0.0771	0.0984	0.0838	0.426	0.388	−0.0068 (−0.0096, −0.0041)

Bias, mean of assigned minus address value; MAE, mean absolute error; r, Pearson correlation with the address value. A, weighted mean; B, weighted centroid; C, PCCF+. Δ_{AC}, mean paired difference in absolute error, weighted mean (A) minus PCCF+ (C) (negative values favour the weighted mean), with a 95% Bayesian bootstrap interval over postal codes (200 replicates). Weighted centroid (B) bias and correlation, and greenness by province: Supplementary Tables S3 and S5. LC, letter carrier; LVR, large volume receiver.

The weighted centroid did not improve the assignment of greenness values, and its MAE was the same as that of PCCF+ overall (paired difference 0.0000, interval −0.0003 to 0.0004), and for six-

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character weighted links (0.0013, −0.0005 to 0.0031), and its correlation for six-character weighted links was lower (0.501). The weighted mean had a smaller mean absolute error than the weighted centroid by 0.0035 overall, by 0.0174 for six-character weighted links, and for every rural delivery mode.

Figure 1 shows an example of a rural postal code in eastern Ontario. The selected postal code has four candidate DA representative points, with EVI values ranging from 0.50 to 0.60, and the largest weight close to the cluster of address locations. The weighted mean (0.537) was higher than the mean address value (0.508), reflecting the greener surroundings of the outlying candidates, but its mean absolute error over the 397 addresses was 0.044, compared with 0.060 for the realized PCCF+ allocation; in this example the weighted centroid, which fell near the addresses, performed slightly better (0.041).



Figure 1. Candidate locations of a rural postal code (six-character weighted link) in eastern Ontario with four candidates and 397 NAR addresses. Circles are PCCF+ candidate DA representative points, sized by weight and shaded by EVI (3 × 3 window, 2025–2026); dots are the reference locations of the addresses; the cross is the weighted centroid (method B). Grey polygons are the candidate DAs. The weighted mean assigns 0.537, the weighted centroid 0.529, and the mean EVI at the addresses is 0.508.

The gain from the weighted mean was concentrated in rural delivery modes in every rurality class (Figure 2). Among rural postal codes, the mean paired difference ranged from −0.011 to −0.019 across rurality classes except the territories (−0.006), and among rural route and suburban service postal codes from −0.009 to −0.019. In the cell-level model, the interaction between delivery mode and rurality was large ($F = 394$ on 21 degrees of freedom), and the adjusted paired difference for rural postal codes ranged from −0.005 in the territories to −0.019 in CMAs. By rurality alone, the paired difference was −0.0012 in CMAs, −0.0031 in CAs and between −0.0102 and −0.0137 in the MIZ classes.

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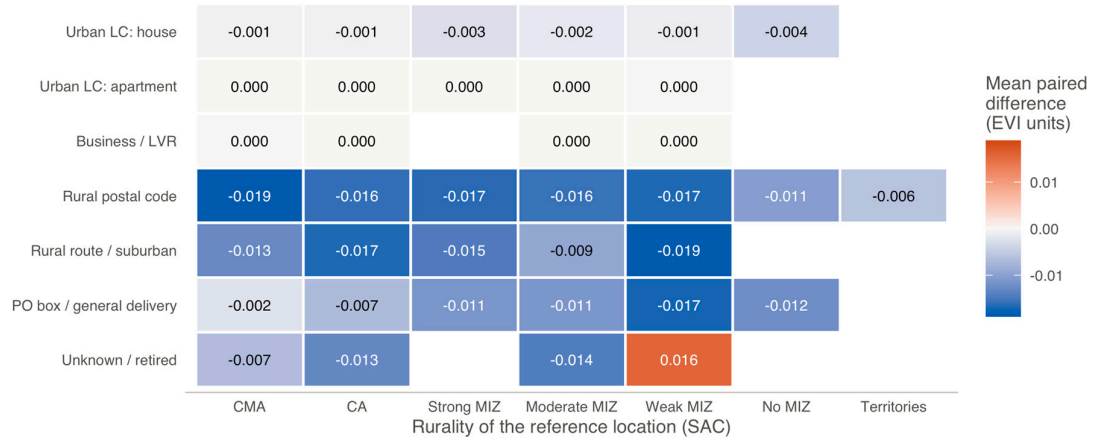
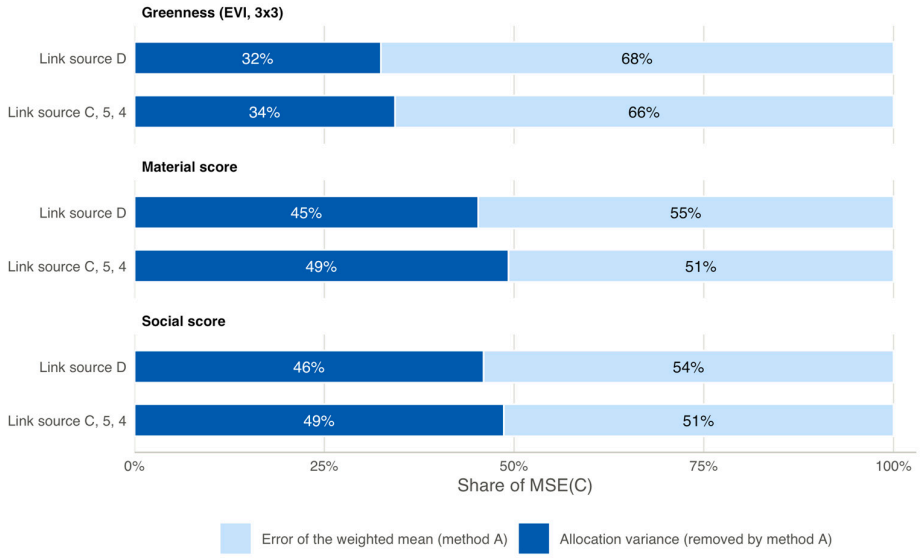


Figure 2. Mean paired difference in absolute EVI error, weighted mean (method A) minus PCCF+ (method C), by delivery mode and rurality of the reference location (EVI 3 × 3 window, 2025–2026; cells with at least 100 addresses). Negative values (blue) favour the weighted mean. The positive cell (unknown or retired postal codes in weak MIZ) contains 108 addresses. LC, letter carrier; LVR, large volume receiver; CMA, census metropolitan area; CA, census agglomeration; MIZ, metropolitan influenced zone.

3.4. Decomposition of the Error of Random Allocation

For six-, five- and four-character weighted links, the MSE of PCCF+ for greenness was 0.01173, and the MSE of the weighted mean plus the mean allocation variance was 0.01172 (relative difference 0.05%, $z = 0.71$); for duplicate links the corresponding values were 0.00104 and 0.00104 (−0.06%, $z = -0.40$). The allocation variance, the part of the error of PCCF+ that the weighted mean removes, accounted for 34% of the MSE of PCCF+ for greenness and 49% for the material deprivation score among six-, five- and four-character weighted links, and for 32% and 45% among duplicate links (Figure 3). The remaining error, the MSE of the weighted mean, comes from the difference between the candidate values and the address, which cannot be removed by averaging over the same candidates.



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Figure 3. Decomposition of the mean squared error of PCCF+ (method C) into the error of the weighted mean (method A) and the mean allocation variance (Equation 8), by measure and link source, under the PCCF+ selection weights. EVI, 3 × 3 window, 2025–2026; scores, MSDI material and social factor scores.

3.5. Area-Based Deprivation

The two quintile rules of the weighted mean behaved differently (Table 6). The quintile of the weighted mean score had the fewest addresses misclassified by two or more quintiles (2.4% overall) but the highest overall misclassification (12.9% for material deprivation, compared with 12.2% for PCCF+), and it assigned fewer addresses to the extreme quintiles. Among addresses with six-character weighted links, 5.7% were assigned the least deprived material quintile, compared with 10.8% at the address.

The modal quintile removed this compression (10.0% assigned to quintile 1) and had the lowest misclassification overall, for six-character weighted and duplicate links, and for every rural delivery mode. Overall, the modal quintile misclassified the material quintile of 10.2% of addresses, 2.0 percentage points (1.9 to 2.1) fewer than PCCF+, with fewer large errors (3.6% vs. 4.6%) and a higher weighted kappa (0.907 vs. 0.887). The reductions were largest for six-character weighted links (36.8% vs. 45.8%, a difference of −9.1 points, −9.4 to −8.8), rural postal codes (37.0% vs. 46.2%) and rural routes and suburban services (33.6% vs. 43.8%). This reduction was also seen for duplicate links (9.0% vs. 10.2%) and urban houses (4.4% vs. 4.9%). Social deprivation showed the same pattern, with overall misclassification of 10.5% for the modal quintile and 12.6% for PCCF+, and 38.8% vs. 48.0% for six-character weighted links. The weighted centroid misclassified slightly more addresses than PCCF+ overall (12.3% vs. 12.2% for material deprivation).

Table 6. Quintile misclassification (%) and weighted kappa for material and social deprivation by method, link source and delivery mode.

Group	Addresses	A mean	A modal	B	C	A modal − C (95% interval)	≥2 off A modal	≥2 off C	κ A modal	κ C
Material deprivation										
All addresses	11,674,361	12.9	10.2	12.3	12.2	−2.0 (−2.1, −1.9)	3.6	4.6	0.907	0.887
Link source										
Unique (F)	6,546,961	1.8	1.8	1.8	1.8	0.0 (0.0, 0.0)	0.7	0.7	0.984	0.984
Six-character weighted (C)	2,190,784	46.8	36.8	46.7	45.8	−9.1 (−9.4, −8.8)	13.4	17.7	0.626	0.524
Duplicate (D)	2,933,912	12.6	9.0	10.3	10.2	−1.3 (−1.4, −1.2)	3.0	3.5	0.915	0.903
Five-character weighted (5)	705	66.8	56.7	35.2	69.1	−12.3 (−27.5, −2.0)	47.9	42.1	0.152	0.165
Four-character weighted (4)	1,999	62.9	64.8	71.5	65.4	−0.6 (−12.0, 12.4)	10.4	19.0	0.328	0.246
Delivery mode										
Urban LC: house	7,637,944	5.8	4.4	4.9	4.9	−0.5 (−0.6, −0.5)	1.5	1.7	0.960	0.955
Urban LC: apartment	1,868,982	2.9	2.9	3.0	2.9	0.0 (−0.1, 0.1)	1.1	1.1	0.974	0.974
Business / LVR	22,657	3.2	3.2	3.4	3.0	0.1 (0.0, 0.3)	2.1	2.0	0.958	0.960
Rural postal code	1,970,391	46.9	37.0	47.1	46.2	−9.1 (−9.5, −8.8)	13.2	17.7	0.621	0.516
Rural route / suburban	143,719	48.5	33.6	43.9	43.8	−10.2 (−11.0, −9.3)	14.5	18.6	0.616	0.506
PO box / general delivery	26,708	50.5	45.9	54.5	50.7	−4.8 (−5.7, −4.0)	22.3	24.3	0.483	0.426
Unknown / retired	3,960	64.0	63.1	62.7	65.6	−2.5 (−9.6, 5.7)	27.1	30.5	0.329	0.273
Social deprivation										
All addresses	11,674,361	13.7	10.5	12.7	12.6	−2.0 (−2.1, −1.9)	3.9	4.8	0.898	0.876

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Link source										
Unique (F)	6,546,961	1.8	1.8	1.8	1.8	0.0 (0.0, 0.0)	0.6	0.6	0.981	0.981
Six-character weighted (C)	2,190,784	50.1	38.8	48.8	48.0	−9.2 (−9.5, −8.8)	14.0	18.5	0.520	0.412
Duplicate (D)	2,933,912	12.9	8.9	10.1	10.1	−1.1 (−1.2, −1.1)	3.5	3.9	0.915	0.904
Five-character weighted (5)	705	52.9	42.4	58.6	46.8	−4.4 (−11.3, 0.7)	31.8	26.1	0.464	0.510
Four-character weighted (4)	1,999	64.2	53.1	55.0	57.1	−4.0 (−16.9, 8.6)	24.6	30.6	0.412	0.359
Delivery mode										
Urban LC: house	7,637,944	6.0	4.4	4.9	4.9	−0.5 (−0.5, −0.4)	1.7	1.9	0.957	0.953
Urban LC: apartment	1,868,982	3.0	2.9	3.1	2.9	0.0 (−0.1, 0.1)	1.1	1.1	0.950	0.950
Business / LVR	22,657	2.9	2.9	2.9	2.8	0.1 (−0.1, 0.5)	1.6	1.7	0.946	0.947
Rural postal code	1,970,391	51.0	39.4	49.7	48.8	−9.4 (−9.7, −8.9)	14.3	19.0	0.505	0.394
Rural route / suburban	143,719	42.3	32.4	41.1	41.0	−8.6 (−9.4, −7.8)	9.7	12.6	0.571	0.467
PO box / general delivery	26,708	53.6	46.7	56.9	53.4	−6.8 (−7.7, −5.7)	17.8	22.2	0.523	0.438
Unknown / retired	3,960	62.1	54.4	58.5	57.4	−3.0 (−9.6, 4.1)	23.9	26.0	0.424	0.400

A mean, weighted mean, quintile of the weighted mean factor score (protocol rule); A modal, weighted mean, modal candidate quintile; B, weighted centroid, quintile of the DA containing the centroid; C, PCCF+. A modal − C, difference in misclassification (percentage points) with a 95% Bayesian bootstrap interval over postal codes (200 replicates). ≥2 off, misclassified by two or more quintiles (%). κ, linearly weighted kappa. LC, letter carrier; LVR, large volume receiver. By province: Supplementary Table S4.

The expected misclassification of PCCF+, $1 - \pi_P(t_i)$, matched its realized misclassification for material deprivation for six-character weighted links (45.87% expected vs. 45.82% observed) and for duplicate links (10.22% for both). Among rural postal codes, use of the modal quintile rule placed more addresses on the diagonal of the misclassification matrix than PCCF+ in every reference quintile. By comparison, the weighted mean score concentrated errors in adjacent quintiles and placed only 37% of addresses in the least deprived quintile to the same quintile (Figure 4). In the logistic model, which compared the weighted mean score rule with the weighted centroid and PCCF+, the interaction of the method with delivery mode and rurality was large (likelihood ratio χ^2 = 7,475 on 62 degrees of freedom). In most urban letter-carrier cells, the weighted mean score rule misclassified more addresses than PCCF+.

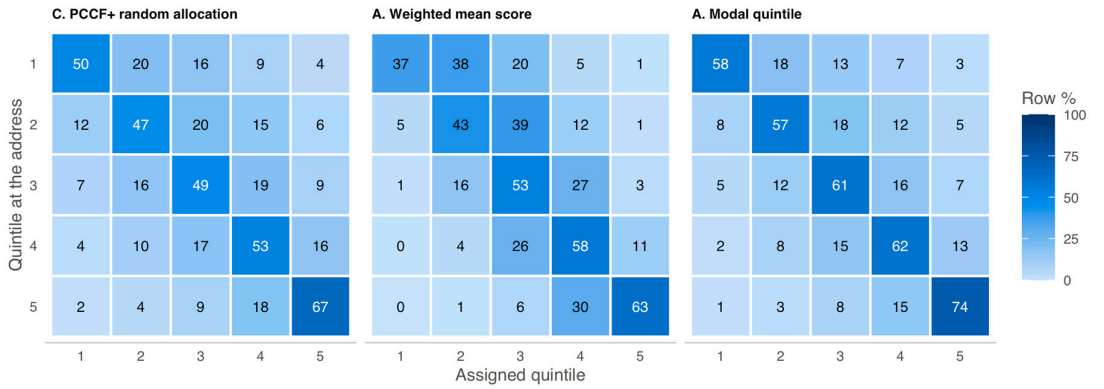


Figure 4. Confusion matrices of the material deprivation quintile for addresses with rural postal codes: PCCF+ (method C), the quintile of the weighted mean score (method A) and the modal candidate quintile (method A). Cells show the percentage of addresses in each reference quintile (row) assigned to each quintile (column).

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3.6. Sensitivity Analyses

Apart from the choice of weights for duplicate links, the sensitivity analyses did not change the pattern of results (Supplementary Tables S6 to S11). With DA population-share weights for duplicate links (sensitivity analysis 1), greenness was essentially unchanged (duplicate-link MAE 0.0152 vs. 0.0151 with equal weights), but the modal quintile misclassified 10.5% of addresses with duplicate links, compared with 9.0% with equal weights and 10.2% for PCCF+, which motivated the change of primary weights (Section 2.3). Weighted centroids calculated from latitude and longitude (sensitivity analysis 2) moved by more than 10 m for 1.5% of addresses and gave identical results at the reported precision. The weighted mean reduced the MAE of PCCF+ by 12.0% for the pixel value, 13.5% for the 3 × 3 window and 14.2% for the 5 × 5 window, and by similar amounts in each year (sensitivity analysis 3), whereas the weighted centroid was within 1% of PCCF+ for every measure. Restriction to addresses with a building reference point (11.7 million addresses; sensitivity analysis 4) and inclusion of urban letter-carrier addresses with a blockface reference point (12.6 million; sensitivity analysis 5) gave the same pattern: greenness MAE 0.0193 for the weighted mean vs. 0.0221 for PCCF+, and 0.0215 vs. 0.0249, respectively, and material misclassification 8.8% for the modal quintile vs. 10.4%, and 10.0% vs. 12.0%. The decomposition held for all nine EVI measures and both MSDI scores, in both link-source groups (all $|z| \leq 1.51$; relative differences ≤ 0.28%; sensitivity analysis 6), with the allocation variance accounting for 29% to 35% of the MSE of PCCF+ for greenness and 45% to 49% for the deprivation scores.

3.7. Reproducible Implementation

The reproducible function was validated using Prince Edward Island under both weight sets, in all 65 compared values (the weighted mean, standard deviation and missing share for nine EVI measures; scores, quintiles, quintile probabilities and modal quintiles for both MSDI dimensions; and the centroid, its DA, EVI, scores and quintiles), with differences below 10^{-9}. The link source inferred from PCCF+ matching agreed with the link source for all PCCF+-coded NAR addresses, including 167 postal codes with three-character weighted links and 11 with two-character links.

4. Discussion

This national comparison separated the two sources of error that are combined in PCCF+ random allocation: the error of the postal code itself, which no method based on the PCCF+ candidates can remove, and the allocation variance added by the random draw, which the weighted mean removes by construction. The allocation variance was a substantial share of the error of PCCF+, a third of the squared error for greenness, and almost half for the deprivation scores among postal codes with weighted candidate sets. The decomposition held to within 0.3% for every measure.

The weighted mean reduced the mean absolute error of greenness by 13.5% overall and by 20% among rural postal codes, with no change in bias, and the modal candidate quintile reduced deprivation quintile misclassification by about 2 percentage points overall and 9 points among rural postal codes. Both use only the PCCF+ files that researchers already hold, require no random seed, and give the same value to every record with the same postal code.

The four hypotheses were largely supported. As expected under the first, the weighted mean and PCCF+ had identical bias and the weighted mean had smaller MAE and RMSE, with gains concentrated in six-character weighted links and rural delivery modes. The second hypothesis was supported only for the modal quintile rule: the quintile of the weighted mean score halved large errors and raised kappa, as anticipated, assigning fewer addresses to the extreme quintiles and increased overall misclassification, because averaging scores across candidates in different quintiles produces intermediate values that none of the candidates has. Classifying an average is not equivalent to averaging classifications, and the modal quintile, which classifies each postal code by the most probable quintile of its candidates, avoids this problem. The third hypothesis was supported: the weighted centroid had the smallest positional error but did not improve greenness,

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and in the least urban classes it had larger errors and lower DA agreement than PCCF+, which is consistent with centroids falling between dispersed settlements. The fourth hypothesis was supported, since, for urban letter carrier postal codes, the methods differed by less than 0.001 EVI units, and the modal quintile rule differed from PCCF+ by 0.5 percentage points or less in misclassification (the weighted mean score rule misclassified about 1 percentage point more for urban houses).

Under the PCCF+ selection probabilities, the weighted mean is the expected value of PCCF+, so in measurement error terms its error relative to the address is the error of a postal-code mean, which is closer to Berkson error, whereas random allocation adds classical-type error with variance V_P [10–12]. The reduction in MSE is a consequence of averaging, and its size can be predicted for any exposure from the weighted variance of the exposure across candidates.

For the primary greenness measure, the predicted and observed MSE agreed to within 0.1%, and the weighted standard deviation of candidate EVI identified where the methods differ, since it was more than ten times larger for rural postal codes than for urban houses. There is a persistent upward bias in greenness for rural postal codes, which was identical for the weighted mean and PCCF+. This shows that averaging cannot correct a systematic difference between the candidate values and the addresses, which may arise in these cases because DA representative points lie in greener surroundings than the dwellings they represent, a mechanism suggested but not tested in the validation study [9]. A related, also untested, possibility is that DA representative points are often far from the road network, whereas addresses follow it.

The poor performance of the weighted centroid for greenness has a practical implication, because the centroid of the candidate points is an intuitive way to assign one location to a postal code and was used in earlier applications of this approach. A centroid minimizes the expected squared distance to the candidates, but the exposure at the centroid is not the average exposure of the candidates, and for rural postal codes with candidates in separate settlements the centroid can fall in farmland, forest or water. Exposure at the centroid may therefore behave more like a value at an arbitrary location than like an average of the candidates, which would explain why its greenness error matched that of PCCF+ while its positional error was lower. Positional accuracy is thus an incomplete guide to exposure accuracy when candidate locations are dispersed.

The sensitivity analysis of the duplicate-link weights (Section 3.6) gave an unexpected result. result for the duplicate-link weights. DA population shares were expected to represent residents better than equal weights, but the duplicate records of a postal code are blockface, block or DA points that may already follow the distribution of its addresses. Population shares may instead give more weight to populous DAs that the postal code only touches, which would explain the higher misclassification. Equal weights are also the PCCF+ selection probabilities, so the decomposition holds exactly for every link source and the weighted mean is the expectation of the PCCF+ result that researchers would otherwise obtain.

This study has several strengths. It used a national address register with building-level reference locations for 12.4 million residential addresses, compared the methods on the same addresses, candidates and exposure layers as the validation study, and verified each step with checks that stopped the analysis if they failed, including an exact empirical test of the decomposition. The portable function was validated and reproduces the PCCF+ link sources for every postal code in the sample.

Several limitations should be considered. Candidate locations for most rural postal codes are DA representative points, so no weighting scheme can recover the position of a dwelling within a DA, and the upward greenness bias, which may arise from the location of those points, was shared by all methods. The weights describe the census population of each candidate DA for a given postal code and not the residential addresses in the NAR, and postal codes and populations may have changed between the census and the NAR release. The NAR mailing postal code was taken as correct, whereas administrative data records may contain outdated and erroneous postal codes. The Berkson-type behaviour of the weighted mean holds only approximately, because the candidate exposures

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are measured at representative points rather than at the homes of residents, and its consequences for exposure-response estimates cannot be tested directly here. Finally, the results apply to exposures that vary at the scale of DAs and MODIS cells; for exposures with sharper gradients, such as traffic-related air pollution, the allocation variance and the gain from averaging may differ, although the decomposition can be computed directly for any exposure.

Future work should examine the effect of the weighted mean and the modal quintile on exposure-response estimates in simulated and real cohorts, apply the methods to other exposures distributed by postal code, and evaluate whether weights derived from address counts in the NAR, rather than census population, further reduce error.

5. Conclusions

Across the 12.4 million Canadian residential addresses analyzed, replacing PCCF+ random allocation with the population-weighted mean of exposure over the PCCF+ candidate locations reduced the mean absolute error of greenness by 13.5% overall and by 20% for rural postal codes without changing bias. Assigning the modal candidate quintile reduced deprivation quintile misclassification by 2 percentage points overall and by 9 points for rural postal codes. The weighted centroid reduced positional error but not exposure error, and the quintile of a weighted mean score compressed assignments toward the middle quintiles. The error of random allocation decomposed exactly into the error of the weighted mean and an allocation variance that the weighted mean removes. The validated R function provided with this study allows both non-random methods to be applied to any dataset that contains residential postal codes, given a licensed copy of PCCF+.

For researchers using data that only contains postal codes as a geographic proxy, the results support three recommendations. First, continuous exposures should be assigned as the population-weighted mean rather than via PCCF+ random allocation, particularly for studies that include rural populations. Second, area-based quintiles should be assigned as the modal candidate quintile rather than the quintile of a weighted mean score, and the quintile probabilities π_p should be retained so that misclassification can be described or corrected. Third, the weighted centroid should not be used as a substitute for the weighted mean. As the weighted mean and PCCF+ differed little for urban letter-carrier postal codes, studies restricted to large urban centres can expect small gains. In contrast, use of the weighted mean for studies of rural health or of rural-urban differences can expect larger gains, of about 20% in the mean absolute error of greenness and 9 percentage points in quintile misclassification among rural postal codes.

Supplementary Materials: The following supporting information is available: Table S1, candidate sets by province and link source; Table S2, positional error by province; Table S3, greenness agreement by province; Table S4, deprivation agreement by province; Table S5, outcomes by link source; Tables S6 to S11, sensitivity analyses 1 to 6; Table S12, cell-level models.

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PCCF+ is distributed by Statistics Canada to licensed users, including through the Data Liberation Initiative and is thus not distributed via the linked repository. Analysis code and the portable R function are available via GitHub at https://github.com/unitepangaea/pccf_exposure_comparison.

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Abbreviations

The following abbreviations are used in this manuscript:

CA	Census agglomeration
CMA	Census metropolitan area
DA	Dissemination area
DMT	Delivery mode type
EVI	Enhanced Vegetation Index
LC	Letter carrier
LVR	Large volume receiver
MAE	Mean absolute error (difference)
MIZ	Census metropolitan influenced zone
MODIS	Moderate Resolution Imaging Spectroradiometer
MSDI	Material and Social Deprivation Index
MSE	Mean squared error
NAR	National Address Register
ODA	Open Database of Addresses
PCCF+	Postal Code Conversion File Plus
RMSE	Root mean square error (difference)
SAC	Statistical Area Classification
WCF	Weighted conversion file

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